5.3.5 Solving with a triangular matrix

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Week 05 due Feb 12, 2025 20:19 IST

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anything with that row. And if you have an appended system, then you don't do

anything with that element on the right in the same row as well. Okay? And then we

look at this and we say ,"Okay, we know that this must be equal to that, but by

the time we get here we know what l_21 is, we know what zeta_1 is, and we know what

L_22 is, and we know what y_2 is." So if we subtract l_21 times zeta_1 from y_2,

then we're left with having to solve L_22 z_2 equal to whatever

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Reading assignment

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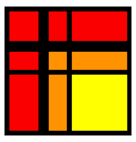

Read Unit 5.3.5 of the notes. [LINK]

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point right here, where we have already overwritten our vector y with z_0. Okay?

In the next step, we would like to compute the next entry of z so that we

have one more entry of z already computed. That entry is zeta_1. So what

we're really saying is when we get to this point, we already know l_10

transpose. We know z_0 and we know psi_1 and we notice that z_1 then is just

equal to, sorry, zeta_1 is

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Homework 5.3.5.1

1/1 point (graded)

Derive an algorithm similar to the one given in Figure 5.3.5.1 for solving U x = z . Update the below skeleton algorithm with the result. (Don't forget to put in the lines that indicate how you "partition and repartition" through the matrix.)

Solve $Ux = z$, overwriting z with x (Variant 1)

$$U \rightarrow \left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right), z \rightarrow \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right)$$

U_{BR} is 0×0 and z_B has 0 elements

while $n(U_{BR}) < n(U)$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \rightarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array}\right)$$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \leftarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \leftarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array}\right)$$

endwhile

Hint: Partition

$$\left(\begin{array}{c|c} U_{00} & u_{01} \\ \hline 0 & \upsilon_{11} \end{array}\right) \left(\begin{array}{c} x_0 \\ \chi_1 \end{array}\right) = \left(\begin{array}{c} z_0 \\ \zeta_1 \end{array}\right).$$

- [x] done

✔

Solution:

For a complete solution, see the notes.

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 5.3.5.2.

1/1 point (graded)

Derive an algorithm similar to the one given in Figure 5.3.5.2 for solving U x = z . Update the below skeleton algorithm with the result. (Don't forget to put in the lines that indicate how you "partition and repartition" through the matrix.)

Solve $Ux = z$, overwriting z with x (Variant 2)

$U \rightarrow \left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right)$, $z \rightarrow \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right)$

U_{BR} is 0×0 and z_B has 0 elements

while $n(U_{BR}) < n(U)$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \rightarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \rightarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array}\right)$$

$$\left(\begin{array}{c|c} U_{TL} & U_{TR} \\ \hline U_{BL} & U_{BR} \end{array}\right) \leftarrow \left(\begin{array}{ccc} U_{00} & u_{01} & U_{02} \\ u_{10}^T & \upsilon_{11} & u_{12}^T \\ U_{20} & u_{21} & U_{22} \end{array}\right), \left(\begin{array}{c} z_T \\ \hline z_B \end{array}\right) \leftarrow \left(\begin{array}{c} z_0 \\ \zeta_1 \\ z_2 \end{array}\right)$$

endwhile

Hint: Partition

$$\left(\begin{array}{c|c} \upsilon_{11} & u_{12}^T \\ \hline 0 & U_{22} \end{array}\right) \left(\begin{array}{c} \chi_1 \\ \hline x_2 \end{array}\right) = \left(\begin{array}{c} \zeta_1 \\ \hline z_2 \end{array}\right).$$

- [x] done

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Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 5.3.5.3

1/1 point (graded)

Let L be an $m \times m$ unit lower triangular matrix. If a multiply and add each require one flop, what is the approximate cost of solving $Lx = y$?

- ( ) $2m^3$
- (•) m^2
- ( ) $2m^2$

Solution:

For complete solution see the notes.

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